

Magnetization in a solenoid

A long solenoid of length $L = 1$ m is made up of $N = 5000$ coils, and is crossed by a current $I = 0.1$ A. The solenoid is filled with a diamagnetic material having magnetic susceptibility $\chi_m = -2 \cdot 10^{-3}$. What is the magnitude of the magnetization vector M ?

- a. $M = 1000$ A/m
- b. $M = 100$ A/m
- c. $M = 0.2$ A/m
- d. $M = 10$ A/m
- ☒ e. $M = 1$ A/m

Magnetizing surface current density

Consider an indefinite solenoid of radius R and number n of coils per unit length, inside which there is a cylinder of magnetic permeability relative μ_r and radius $r < R$, coaxial to the solenoid. If I is the conduction current in the solenoid, the magnitude of the magnetizing surface current density j_{sm} is:

- a. $j_{sm} = 0$
- b. $j_{sm} = \mu_0(\mu_r - 1)nI$
- c. $j_{sm} = \mu_0(\mu_r - 1)nI/\pi r^2$
- d. $j_{sm} = \mu_0(\mu_r - 1)nI\pi r^2$
- ☒ e. $j_{sm} = (\mu_r - 1)nI$

Curie-Weiss law for ferromagnets

Curie's law for a ferromagnet states that magnetic susceptibility is

a. $\chi_m = C_P/(T - T_C)$ for $T > T_C$

b. $\chi_m = C_P/T$

c. $\chi_m = C_P/(T - T_C)$ for $T < T_C$

d. $\chi_m = C_P/(T - T_C)$

Magnetic Field in a Solenoid with Ferromagnetic Core

A straight solenoid long 10 m, with a radius of 10 cm, consists of 1000 turns wound on a ferromagnetic core with relative magnetic permeability equal to 100. If a current of 10 A flows in the windings, the magnetic field $\mathbf{B}$ inside the solenoid is approximately equal to:

- a. $B \approx 6 \text{ T}$
- b. $B \approx 0.83 \text{ T}$
- ☒ c. $B \approx 1.3 \text{ T}$
- d. $B \approx 0.6 \text{ T}$
- e. $B \approx 2.6 \text{ T}$

Temperature Dependence of Diamagnetism

In a diamagnet the magnetic susceptibility

- a. depends on the applied magnetic field
- b. grows with temperature
- c. decreases with temperature
- ☒ d. does not vary with temperature

Effect of diamagnetic material on B field

In the presence of a diamagnetic material in a long and narrow solenoid, with equal conduction current:

- a. the field $\mathbf{B}$ does not change
- ☒ b. the field $\mathbf{B}$ decreases
- c. if $\mathbf{B}$ increases or decreases depends on the type of diamagnetic material
- d. the field $\mathbf{B}$ increases

Material type based on susceptibility

Question: If $\mathbf{M} = \chi_m(T)\mathbf{H}$ the material is

- a. ferromagnetic because $\chi_m(T)$ doesn't depend from $\mathbf{H}$
- b. diamagnetic
- ☒ c. paramagnetic
- d. diamagnetic and paramagnetic

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Definition of surface amperian current

The product $\mathbf{M} \times \mathbf{n}$ where $\mathbf{M}$ is the magnetization vector of a material and $\mathbf{n}$ is the unit vector normal to surface of the material is equal to

- a. normal component of vector $\mathbf{M}$
- b. volume density of amperian currents
- ☒ c. surface density of amperian currents
- d. zero because $\mathbf{M}$ and $\mathbf{n}$ are always parallel

Solenoidal Nature of $\mathbf{H}$

In an isotropic material $\mathbf{B} = \mu\mathbf{H}$. This means that $\mathbf{H}$ is always solenoidal.

- ☒ a. No
- ☐ b. Yes
- ☐ c. Yes but just in the paramagnetic and ferromagnetic materials.
- ☐ d. Yes but just in a diamagnetic material.

Definition of bulk amperian current

The bulk density of amperian currents $\mathbf{j}_m$ in a material with magnetization vector $\mathbf{M}$, is equal to

- a. $\mathbf{j}_m = \text{rot } \mathbf{M}$
- b. $\mathbf{j}_m = \mathbf{M} \times \mathbf{n}$
- c. $\mathbf{j}_m = \text{div } \mathbf{M}$
- d. $\mathbf{j}_m = \text{grad } \mathbf{M}$

Amperian currents inside a magnetized material

If the magnetization vector $\mathbf{M}$ in a material is equal to $\mathbf{M}(x, y, z) = ax\mathbf{i} + by\mathbf{j} + cz\mathbf{k}$, where a, b and c are constants of suitable size, the amperian currents inside the material:

- a. depends from the values of a, b and c
- b. The question is wrong: the amperian currents don't depends from $\mathbf{M}$
- ☒ c. are zero
- d. are different from zero

Characteristics of the Magnetizing Field $\mathbf{H}$

Which of the following property is characteristic of magnetizing field $\mathbf{H}$?

- a. Its value does not depend on the possible presence of magnetization currents
- b. Its circuitry along a closed line does not depend on the possible presence of conduction currents concatenated to the same line
- ☒ c. Its circuitry along a closed line does not depend on the possible presence of magnetization currents concatenated to the same line
- d. It is solenoidal

Boundary Conditions at Magnetic Interfaces

The ratio of the tangential components B_{1t} and B_{2t} of the magnetic induction field $\mathbf{B}$ measured at the interface between two materials having $\chi_{1m} = 0.3$ and $\chi_{2m} = 0.5$ respectively is:

☒ a. $B_{1t}/B_{2t} = 0.87$

b. $B_{1t}/B_{2t} = 1.67$

c. $B_{1t}/B_{2t} = 1.15$

d. $B_{1t}/B_{2t} = 0.60$

Magnetization current density in a cylinder

A cylinder is magnetized in a non-uniform way. The magnetization inside is $\mathbf{M} = by\mathbf{u}_x - bx\mathbf{u}_y + bz\mathbf{u}_z$ (where b is a constant of suitable size). What is the current density of magnetization $\mathbf{j}_m$ inside?

a. $\mathbf{j}_m = 2b\mathbf{u}_z$

b. $\mathbf{j}_m = \mathbf{0}$

☒ c. $\mathbf{j}_m = -2b\mathbf{u}_z$

d. $\mathbf{j}_m = b\mathbf{u}_x - b\mathbf{u}_y$

Bound currents for uniform magnetization

In a sample of paramagnetic material there is a magnetization vector $\mathbf{M} = M\mathbf{k}$ where M is a constant of suitable size. What are the Amperian current densities of volume $\mathbf{j}_v$ and surface $\mathbf{j}_s$?

- a. $\mathbf{j}_v = 0$, $\mathbf{j}_s$ is indeterminate if the sample shape is not specified
- b. $\mathbf{j}_v = \mathbf{M}$, $\mathbf{j}_s = 0$
- c. $\mathbf{j}_v = 0$, $\mathbf{j}_s = \mathbf{M}$
- d. $\mathbf{j}_v = 0$, $\mathbf{j}_s = 0$

Relative magnetic permeability of a material

The magnetic field and magnetization within a homogeneous and isotropic paramagnetic material are valid in module respectively $B = 8 \cdot 10^{-4}$ T and $M = 200$ A/m. What is the relative magnetic permeability μ_r of the material?

☒ a. $\mu_r = 1.46$

b. $\mu_r = 1.31$

c. $\mu_r = 3.18$

d. $\mu_r = 4.18$

Definition of the Magnetization Vector

The magnetization vector of a substance indicates:

- a. the magnetic moment of the substance per unit of volume
- b. the magnetic moment of the substance
- c. the magnetic moment of the individual atoms / molecules of the substance
- d. the magnetic field present within the substance

Magnetic field of a concentric half-coil circuit

The circuit shown in the figure is constituted by two concentric circular half coils of radius respectively $r_1 = 10$ cm and $r_2 = 20$ cm. The circuit is carrying a current $i = 2$ A and is immersed in a medium having a magnetic permeability $\kappa_m = 1.1$. Determine the value of the modulus of the magnetic field B generated by the circuit at the point P, the center of the two half coils.

a. $B = 1.037 \cdot 10^{-4}$ T

b. $B = 2.074 \cdot 10^{-5}$ T

☒ c. $B = 1.037 \cdot 10^{-5}$ T

d. $B = 1.037 \cdot 10^{-3}$ T

Relationship between magnetic fields in matter

Which of the following equations expresses the relationship between the magnetization $\mathbf{M}$, the magnetic field $\mathbf{B}$ and the magnetizing field $\mathbf{H}$ in a paramagnetic material?

☒ a. $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$

b. $\mathbf{B} = 1/\mu_0(\mathbf{H} + \mathbf{M})$

c. $\mathbf{H} = \mu_0(\mathbf{B} + \mathbf{M})$

d. $\mathbf{M} = \mu_0(\mathbf{B} + \mathbf{H})$

Magnetization current density in a cylinder 2

A cylinder is magnetized in a non-uniform way. The magnetization inside is $\mathbf{M} = -ay\mathbf{u}_x - bz\mathbf{u}_y + cz\mathbf{u}_z$ (where a, b , and c are constants). What is the current density of magnetization inside?

a. $\mathbf{j}_m = c\mathbf{u}_z$

b. $\mathbf{j}_m = b\mathbf{u}_x + a\mathbf{u}_z$

c. $\mathbf{j}_m = c\mathbf{u}_x - b\mathbf{u}_y$

d. $\mathbf{j}_m = -ay\mathbf{u}_x - bz\mathbf{u}_y + cz\mathbf{u}_z$

Magnetization inside a current-carrying wire

A wire conductor of infinite length with radius R and relative magnetic permeability κ_m is traversed by a current i . Determine the value of the modulus of the magnetization of the material at a point P within the wire and away from its axis r ($r < R$).

- a. $M = i/(2\pi r)$
- b. $M = \kappa_m \mu_0 i/(2\pi r)$
- c. $M = \kappa_m \mu_0 i r/(2\pi R^2)$
- d. $M = (\kappa_m - 1) i r/(2\pi R^2)$